

Gated Dual Attention Unit Neural Networks for Remaining Useful Life Prediction of Rolling Bearings

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Abstract—In the mechatronic system, rolling bearing is a frequently used mechanical part, and its failure may result in serious accident and major economic loss. Therefore, the remaining useful life (RUL) prediction of rolling bearing is greatly indispensable. To accurately predict the RUL of the rolling bearing, a new kind of gated recurrent unit neural network with dual attention gates, namely, gated dual attention unit (GDAU), is proposed. With the acquired life-cycle vibration data of a rolling bearing, a series of root mean squares at different time instants are calculated as the health indicator (HI) vector. Next, the to-be HI sequence is predicted by GDAU according to the existing HI vector, and then the RUL of the rolling bearing is estimated. The experimental results show that the proposed GDAU can effectively predict the RULs of rolling bearings, and it has higher prediction accuracy and convergence speed than the conventional prediction methods.

Index Terms—Attention gate, gated recurrent unit (GRU), health indicator (HI), life prediction, long-term prediction.

NOMENCLATURE

A	Cell neuron of GRU.
h_t	Hidden state at time t .
h_{t-1}	Hidden state at time $t-1$.
σ	Logistic sigmoid function.
$\tanh$	Hyperbolic tangent function.
r_t	Output of reset gate.
z_t	Output of update gate.
U, W	Weight matrices.
b	Bias matrix.
x_t	Input data at time t .
$\hat{h}_t$	Temporary hidden state at time t .
X	Input data of attention mechanism.
z	Index of selected input data.
q	Task-related query vector.

α_i	Attention distribution of the i th input data.
$s(x_i, q)$	Attention scoring function.
V	Additive parameter related to the attention gate.
$\hat{h}_t$	Output of attention gate.
$\hat{A}_t$	Ratios of attention.
$\hat{A}_t$	Candidate attention value.
A_t	Attention distribution.
S	Training vector with m samples.
Y	Normalized vector of training vector.
W	Reconstruct matrix.
k	Cell number of input layer.
$\bar{x}$	Predicted data.
x'	Actual data.
Rul	Actual remaining useful life.
$\bar{Rul}$	Estimated remaining useful life.
Er_i	Percent error for the i -th experiment.
A_i	Score of accuracy in the i -th experiment.
MAE	Mean absolute error.
NRMSE	Normalized root mean square error.
MAPE	Mean absolute percentage error.
RMSE	Root mean square error.

I. INTRODUCTION

AS ONE of the most critical components, rolling element bearing is widely used in various mechatronic systems. Under the complex working condition and harsh environment, different kinds of rolling element bearing failures would result the industrial equipment in abnormal state, such as accuracy loss, productivity reduction, the increase of downtime, and even the rise of safety risks [1]–[4]. To avoid these problems effectively, the prediction of bearing remaining useful life (RUL) is regarded as the most major task, and it is greatly beneficial to industrial production.

Most of RUL prediction methods can be categorized into physics-based and data-driven methods [5], [6]. Physics-based methods are used to describe the degradation process of a machine or component by setting up physical models, and then predict the RUL [7]. However, the construction of physics-based models may be hindered by the complex system structure and working condition. Moreover, the physical model is just able to accurately represent the degradation trend in some situations. With the rapid development of sensor technology, computer science, and artificial intelligence theory, data-driven

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methods have become more and more attractive in prognostics and health management (PHM) [8]. In various data-driven methods, machine-learning technology is regarded as a powerful and efficient solution, which is applied to adaptively model the degradation process by using measured data. As one of the advanced machine-learning technologies, deep learning has been widely used for the RUL prediction [9], [10]. Guo *et al.* [11] proposed a recurrent neural network based on health indicator (RNN-HI) to predict the RUL of rolling element bearing, where the most sensitive features from six-related similarity and eight classical time-frequency features were used as the input of RNN. Chen *et al.* [12] proposed the frozen convolution and activated memory network for RUL estimation of bearing, and raw vibration signals were taken as the input. Zhu *et al.* [13] built a multiscale convolution neural network (CNN) to predict the RUL of bearing, which kept the global and local information synchronously compared to a traditional CNN. Ren *et al.* [14] proposed a new deep learning-based prediction framework for bearing RUL estimation by utilizing deep autoencoder and deep neural network (DNN). Li *et al.* [15] proposed a deep generative adversarial network (DGAN) for machinery RUL prediction. Liu *et al.* [16] proposed a joint-loss CNN (JL-CNN) that could implement the fault recognition and RUL estimation in parallel. Li *et al.* [17] proposed a deep-learning method based on multiscale feature extraction for RUL estimation.

As a connectionist model, RNN is able to selectively pass information across sequential steps, and process one element of sequential data at a time, which is unlike standard feedforward neural network [18]. Owing to this merit, RNN is highly suitable for processing sequence information. However, RNN possesses the problems of vanishing gradient and gradient explosion. In order to overcome these problems, long short-term memory (LSTM) was proposed by Hochreiter and Schmidhuber in 1997 [19], which introduced input gate, output gate, and cell state into RNN. Then, the gated recurrent unit (GRU) was proposed by Cho *et al.* in 2014 [20], that could be considered as an improved LSTM-architecture network. LSTM and GRU have been widely applied to RUL prediction and fault diagnosis. Hinch and Tkouat [21] proposed the convolutional LSTM to estimate the RUL of rolling bearing. Yuan *et al.* [22] built LSTM neural network for fault diagnosis and RUL estimation of aero-engine in the case of complicated conditions with strong noise. Malhotra *et al.* [23] adopted LSTM encoder-decoder to obtain health index of multisensor time-series data, which was then utilized to predict the machine RUL. Huang *et al.* [24] built bidirectional LSTM neural network for RUL prediction of engineered systems with multiple working conditions. Jiang *et al.* [25] combined CNN with LSTM neural networks for RUL prediction of bearings. Chen *et al.* [26] proposed a method based on GRU neural network for predicting the nonlinear deterioration process. Ren *et al.* [27] proposed a method based on multiscale dense GRU neural network for the RUL prediction of bearing.

Attention mechanism is actually a resource allocation scheme that can solve the problem of information overload and distribute computing resource to significant information. Attention mechanism has been applied to image identification [28], natural language processing [29], etc. In the field of prediction, researchers

have focused on attention mechanism. Qin *et al.* [30] proposed a dual-stage attention-based RNN for time series prediction, which was superior to the state-of-the-art methods. Fernando *et al.* [31] combined the soft and hardwired attention to build an LSTM framework to predict human trajectory and detect abnormal event. Wu *et al.* [32] embedded attention mechanism into LSTM to ensure LSTM have a strong attention capability and proposed a context-aware attention LSTM (CA-LSTM) network to predict flood. Qin *et al.* [33] proposed a new type of long short-term memory with macro-micro attention (MMALSTM) for gear and bearing RUL prediction.

RNN only remembers short-term memory, but GRU has the ability to remember long-term memory as a result of the gate mechanism, which solves the problem of long-term dependency. However, it does not pay much attention to the significant information in the process of prediction. Fortunately, the attention mechanism is able to pay close attention to the important information and suppress insignificant information. Therefore, this article designs a new attention gate according to the structural feature of GRU, and then this attention gate combined with another attention gate are effectively integrated into the interior of GRU. The newly designed gate has the ability to focus on the output information of the reset and update gates, while the other attention gate is applied to pay attention to the information of input data and hidden state at the previous time instant. The two attention gates can enhance the ability of GRU to solve the problem of long-term dependency and reduce much computing resource. The new development of GRU is called gated dual attention unit (GDAU) in this study.

Using the life-cycle vibration data set of rolling bearing, the root mean square (RMS) of bearing vibration signal is calculated as the HI. With the RMS index, a GDAU-based RUL prediction approach for rolling bearing is proposed in this article. The experiment results show that GDAU can really outperform GRU and its variants in the rolling bearing RUL prediction. Last, we summarize the main contributions of this work as follows.

- 1) A new attention gate is designed via the structure of GRU, which can select the important output information of the reset and update gates to update the hidden state. With this new attention gate, gated attention unit (GAU) neural network is first proposed.
- 2) By integrating two attention gates, GDAU neural network is first constructed, which can pay attention to the significant information of input data, hidden state, and the outputs of the reset and update gates simultaneously. Thus, it has good long-term and short-term predictive performance.
- 3) Based on RMS index and GDAU, a new methodology for predicting the RULs of rolling bearings is proposed, which can estimate the bearing RUL under small sample. Moreover, it has higher prognosis accuracy and convergence speed than other current methods.

The rest of this article is organized as follows. Section II introduces GRU and attention mechanism. Section III introduces a novel GRU neural network with dual attention gates, named GDAU, and the detailed construction process is explained. In

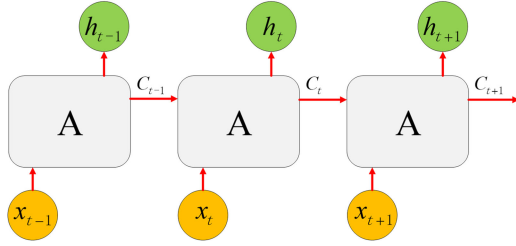


Fig. 1. Whole GRU neural network structure.

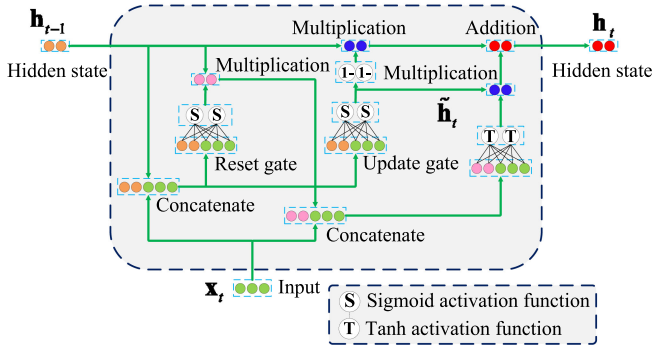


Fig. 2. Structure of a single cell neuron A.

Section IV, some comparative experiments are conducted on the bearing data set to show the advantage of the proposed methodology. Finally, Section V concludes this article.

II. BASIC THEORETICAL BACKGROUND

A. Gated Recurrent Unit

GRU is a development of traditional RNN. A gate mechanism is added to the structure of GRU, and different gates can process information differently. The whole structure of the GRU neural network is shown in Fig. 1.

The chain structure of the GRU neural network is formed by the interconnection of some single cell neurons. The cell neuron A consists of reset and update gates. The specific structure of a single cell neuron A is shown in Fig. 2.

The output hidden state $\mathbf{h}_t$ of a single cell neuron A at time t is calculated by

$$\mathbf{r}_t = \sigma(\mathbf{U}_r \mathbf{x}_t + \mathbf{W}_r \mathbf{h}_{t-1} + \mathbf{b}_r) \quad (1)$$

$$\mathbf{z}_t = \sigma(\mathbf{U}_z \mathbf{x}_t + \mathbf{W}_z \mathbf{h}_{t-1} + \mathbf{b}_z) \quad (2)$$

$$\tilde{\mathbf{h}}_t = \tanh(\mathbf{U}_h \mathbf{x}_t + \mathbf{W}_h (\mathbf{r}_t \odot \mathbf{h}_{t-1}) + \mathbf{b}_h) \quad (3)$$

$$\mathbf{h}_t = (1 - \mathbf{z}_t) \odot \mathbf{h}_{t-1} + \mathbf{z}_t \odot \tilde{\mathbf{h}}_t \quad (4)$$

where σ and $\tanh$ are, respectively, the logistic sigmoid and hyperbolic tangent function; $\mathbf{r}_t$ is the output of reset gate; $\mathbf{z}_t$ is the output of update gate; $\mathbf{U}_r$, $\mathbf{W}_r$, $\mathbf{U}_z$, $\mathbf{W}_z$, $\mathbf{U}_h$, and $\mathbf{W}_h$ are weight matrices; $\mathbf{b}_r$, $\mathbf{b}_z$, and $\mathbf{b}_h$ are bias matrixes; $\mathbf{x}_t$ represents the input data at the time instant t ; $\mathbf{h}_{t-1}$ represents the hidden state at time $t - 1$; $\tilde{\mathbf{h}}_t$ represents the temporary hidden state at time t ; and $\odot$ represents dot product.

B. Attention Mechanism

Attention is a complex cognitive ability, which is indispensable to human. It refers to the ability of people to focus on some important information but ignore unimportant information. Like the attention mechanism of human, we can make the neural network only select some key information for processing to enhance its learning ability. Suppose that $\mathbf{x} = [x_1, \dots, x_n]$ denotes input data, where n is the data length. It is not necessary for inputting all data into the neural network to compute, and the better way is to select some task-related information from $\mathbf{x}$ for the input of the neural network, which can effectively save computing resource. We use the attention variable $z \in [1, n]$ to indicate the index of selected input data. If $z = i$, it means the i th input data is selected. Given the task-related query vector q and input data $\mathbf{x}$, the attention distribution α_i of the selecting i th input data can be calculated by

$$\alpha_i = \text{softmax}(s(x_i, q)) = \frac{\exp(s(x_i, q))}{\sum_{i=1}^n \exp(s(x_i, q))} \quad (5)$$

where $s(x_i, q)$ is the attention scoring function, and it is defined as follows:

$$s(x_i, q) = \mathbf{V}^T \tanh(\mathbf{W} x_i + \mathbf{U} q) \quad (6)$$

where $\mathbf{V}$, $\mathbf{W}$, and $\mathbf{U}$ are network parameters learnt in the training of the neural network. And their specific meanings can be seen in the next section.

III. ATTENTION GATES IN GRU

A. Recurrent Attention Unit

On the basis of attention mechanism, Zhong *et al.* [34] introduced an attention gate into the traditional GRU, and constructed a new neural network called recurrent attention unit (RAU), which can be regarded as a variant of the GRU. The detailed structure of RAU is illustrated in Fig. 3.

From Fig. 3, it can be seen that RAU consists of reset gate, update gate and attention gate. It is computed by the following equations:

$$s(\mathbf{x}_t, \mathbf{h}_{t-1}) = \mathbf{V}^T \tanh(\mathbf{W}_s \mathbf{x}_t + \mathbf{U}_s \mathbf{h}_{t-1}) \quad (7)$$

$$\alpha_t = \text{softmax}(s(\mathbf{x}_t, \mathbf{h}_{t-1})) = \frac{\exp(s(\mathbf{x}_t, \mathbf{h}_{t-1}))}{\sum_{i=1}^n \exp(s(\mathbf{x}_t, \mathbf{h}_{t-1}))} \quad (8)$$

$$\hat{\mathbf{h}}_t = \tanh(\alpha_t \odot (\mathbf{W}_{\hat{h}} \mathbf{x}_t + \mathbf{U}_{\hat{h}} \mathbf{h}_{t-1})) \quad (9)$$

$$\mathbf{h}_t = (1 - \mathbf{z}_t) \odot \mathbf{h}_{t-1} + (\mathbf{z}_t \odot \tilde{\mathbf{h}}_t)/2 + (\mathbf{z}_t \odot \hat{\mathbf{h}}_t)/2 \quad (10)$$

where $\mathbf{V}$ is additive parameter related to the attention gate; $\mathbf{W}_s$ and $\mathbf{U}_s$ are weight matrices related to the scoring function; $\mathbf{W}_{\hat{h}}$ and $\mathbf{U}_{\hat{h}}$ are weight matrices related to the output of attention gate $\hat{\mathbf{h}}_t$; and α_t denotes the attention distribution vector.

B. Gated Attention Unit

Both the reset and update gates are used for updating the hidden state at time t . The reset gate can discard the information of the hidden state at time $t-1$ that has nothing to do with the future state, and it is mainly used to process short-term

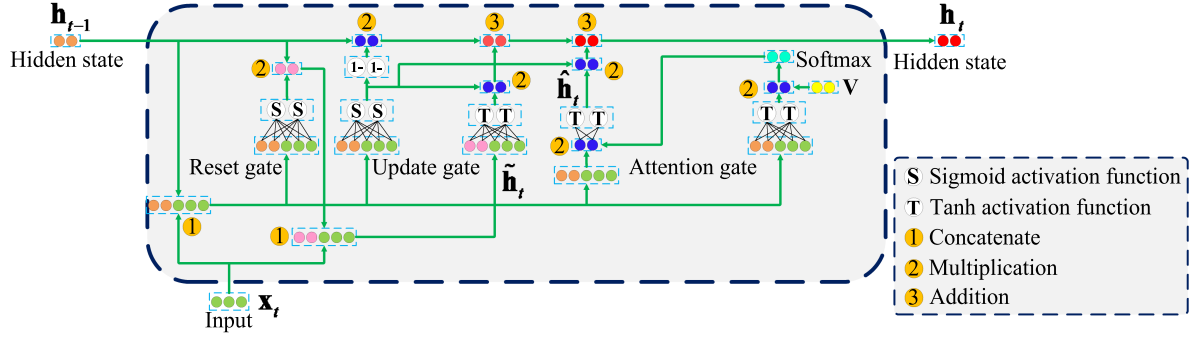


Fig. 3. Recurrent Attention Unit (RAU).

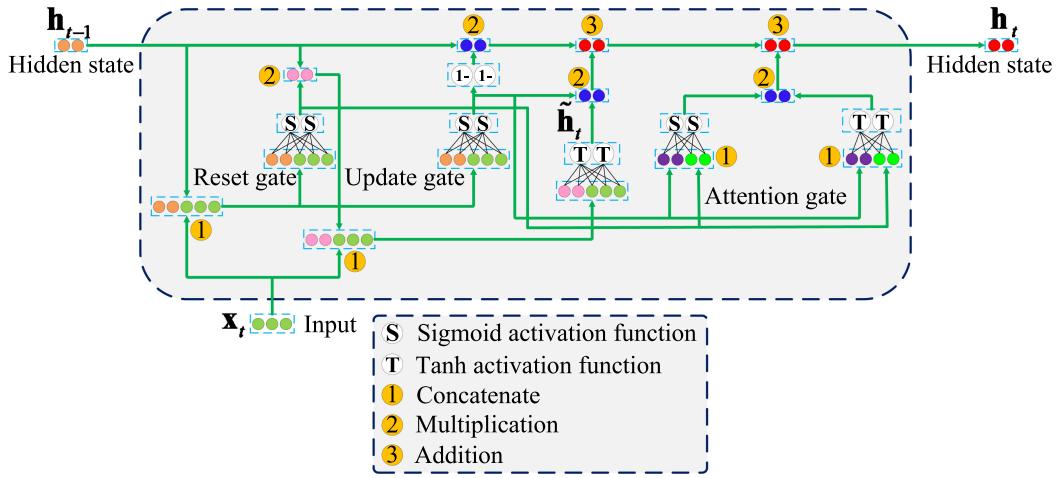


Fig. 4. Gated attention unit (GAU).

information; while the update gate is able to update the hidden state at time t by the input data at time t and the hidden state at time $t-1$, and it principally handles long-term information. However, the output information of the reset and update gates is not paid attention, which may result in that some significant information related to the future trend is lost. Therefore, by embedding attention mechanism into GRU structure, we propose a new variant of GRU, namely, GAU, which can retain the valuable output information of the reset and update gates.

The structure of GAU is illustrated in Fig. 4. It is similar to that of RAU. The difference is that the information of the input data and the hidden state at the previous moment is paid attention in RAU, while the outputs of the reset and update gates that contain the short- and long-term information are paid attention in GAU. The calculation equations of attention gate in GAU are defined as follows:

$$\tilde{\mathbf{A}}_t = \sigma(\mathbf{W}_{\tilde{\mathbf{A}}} \mathbf{r}_t + \mathbf{U}_{\tilde{\mathbf{A}}} \mathbf{z}_t + \mathbf{b}_{\tilde{\mathbf{A}}}) \quad (11)$$

$$\hat{\mathbf{A}}_t = \tanh(\mathbf{W}_{\hat{\mathbf{A}}} \mathbf{r}_t + \mathbf{U}_{\hat{\mathbf{A}}} \mathbf{z}_t + \mathbf{b}_{\hat{\mathbf{A}}}) \quad (12)$$

$$\mathbf{A}_t = \tilde{\mathbf{A}}_t \odot \hat{\mathbf{A}}_t \quad (13)$$

where $\mathbf{W}_{\tilde{\mathbf{A}}}$, $\mathbf{U}_{\tilde{\mathbf{A}}}$, $\mathbf{W}_{\hat{\mathbf{A}}}$, and $\mathbf{U}_{\hat{\mathbf{A}}}$ are weight matrixes, and $\mathbf{b}_{\tilde{\mathbf{A}}}$, and $\mathbf{b}_{\hat{\mathbf{A}}}$ are bias matrixes. $\tilde{\mathbf{A}}_t$ is computed by the sigmoid function σ , which represent the ratios of attention on the outputs of the reset gate $\mathbf{r}_t$ and update gate $\mathbf{z}_t$. Via the Tanh function, the candidate attention value $\hat{\mathbf{A}}_t$ is calculated, which determines whether the information is positive or negative. $\mathbf{A}_t$ represents the attention distribution on $\mathbf{r}_t$ and $\mathbf{z}_t$.

The GAU also has the reset gate, update gate and the temporary hidden state, which can be respectively calculated by Eqs. (1)–(3). Through Eqs. (1)–(3) and (11)–(13), the hidden state at time t is reformulated as

$$\mathbf{h}_t = (1 - \mathbf{z}_t) \odot \mathbf{h}_{t-1} + (\mathbf{z}_t \odot \tilde{\mathbf{h}}_t) + \mathbf{A}_t \quad (14)$$

The advantage of GAU is that it focuses on the important output information of the two gates and automatically discard some unimportant information. Therefore, it can learn more useful dependencies with respect to the unknown data, so as to enhance the ability of GRU.

C. Gated Dual Attention Unit

From the above, we can know that GAU and RAU can pay attention to different crucial information via different attention

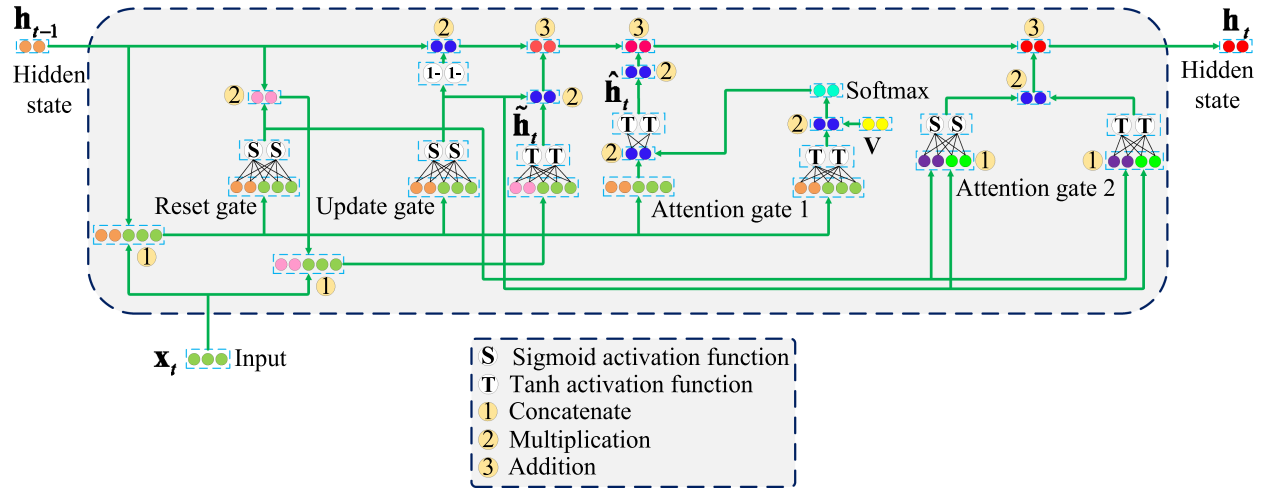


Fig. 5. Gated dual attention unit (GDAU).

gates, which are formed by different data flows. However, they are only able to focus on some useful information separately. In order to further improve the predictive ability of GRU, both attention gates are simultaneously integrated in GRU, then a novel variant of GRU is proposed, which is named as GDAU. The advantages of RAU and GAU are combined in GDAU. It uses not only the important information from the input data and the hidden state, but also the significant information from the outputs of the reset and update gates to update the hidden state. Compared with GRU, RAU, and GAU, the proposed GDAU improves the ability the ability of long-term prediction.

The structure of GDAU is shown in Fig. 5. Obviously, GDAU has four gates, that is, reset gate, update gate, the proposed attention gate 1, and attention gate 2. According to the structure of GDAU, all the transaction equations of GDAU can be formulated as

$$\begin{aligned}
 \mathbf{r}_t &= \sigma(\mathbf{U}_r \mathbf{x}_t + \mathbf{W}_r \mathbf{h}_{t-1} + \mathbf{b}_r) \\
 \mathbf{z}_t &= \sigma(\mathbf{U}_z \mathbf{x}_t + \mathbf{W}_z \mathbf{h}_{t-1} + \mathbf{b}_z) \\
 \tilde{\mathbf{h}}_t &= \tanh(\mathbf{U}_h \mathbf{x}_t + \mathbf{W}_h (\mathbf{r}_t \odot \mathbf{h}_{t-1}) + \mathbf{b}_h) \\
 \mathbf{s}(\mathbf{x}_t, \mathbf{h}_{t-1}) &= \mathbf{V}^T \tanh(\mathbf{W}_s \mathbf{x}_t + \mathbf{U}_s \mathbf{h}_{t-1}) \\
 \boldsymbol{\alpha}_t &= \text{softmax}(\mathbf{s}(\mathbf{x}_t, \mathbf{h}_{t-1})) = \frac{\exp(\mathbf{s}(\mathbf{x}_t, \mathbf{h}_{t-1}))}{\sum_{i=1}^n \exp(\mathbf{s}(\mathbf{x}_t, \mathbf{h}_{t-1}))} \\
 \hat{\mathbf{h}}_t &= \tanh(\boldsymbol{\alpha}_t \odot (\mathbf{W}_{\hat{h}} \mathbf{x}_t + \mathbf{U}_{\hat{h}} \mathbf{h}_{t-1})) \\
 \tilde{\mathbf{A}}_t &= \sigma(\mathbf{W}_{\tilde{A}} \mathbf{r}_t + \mathbf{U}_{\tilde{A}} \mathbf{z}_t + \mathbf{b}_{\tilde{A}}) \\
 \hat{\mathbf{A}}_t &= \tanh(\mathbf{W}_{\hat{A}} \mathbf{r}_t + \mathbf{U}_{\hat{A}} \mathbf{z}_t + \mathbf{b}_{\hat{A}}) \\
 \mathbf{A}_t &= \tilde{\mathbf{A}}_t \odot \hat{\mathbf{A}}_t \\
 \mathbf{h}_t &= (1 - \mathbf{z}_t) \odot \mathbf{h}_{t-1} + (\mathbf{z}_t \odot \tilde{\mathbf{h}}_t)/2 + (\mathbf{z}_t \odot \hat{\mathbf{h}}_t)/2 + \mathbf{A}_t
 \end{aligned} \tag{15}$$

From (15), we can see that more valuable information is contained in the hidden state $\mathbf{h}_t$. Therefore, the GDAU neural network has a fine predictive ability, even under the condition of fewer training samples.

The four gates have different roles. The reset gate is used to determine how much information will be dropped, which is also the input of attention gate 2. The update gate is able to control how much information of the input data and hidden state at

the previous moment will be used for updating the hidden state at the current moment. Both the outputs of reset and update gates are applied to the input of attention gate 2. The attention gate 1 is mainly used to process the input data and the hidden state for learning the key information on the dependency. The attention gate 2 can also learn the evolution law of the data from the output information of reset and update gates. It could compute the ratios of attention on information and distinguish the information is positive or negative. Especially, owing to the two attention gates, the predictive performance of the proposed GDAU neural network is greatly improved.

D. RUL Prediction Approach for the Rolling Bearing

In order to predict the RUL of rolling bearing, an effective indicator to evaluate the degradation trend of bearing is required. It has been found that the time domain feature can reflect the overall health of the bearing [35]. And the RMS index is one of the most widely used time domain features, and it has been successfully applied to the bearing RUL prediction [36]. Therefore, we use the RMS of vibration signal as the HI to estimate the RUL of rolling element bearing. The predictive procedure of bearing RUL is illustrated in Fig. 6. The steps of the bearing RUL prediction approach based on GDAU neural network are summarized as follows.

- 1) During the life-cycle of bearings, collect the vibration signals. The sampling time is T and the interval between two adjacent samples is T_s . Suppose the number of samples is n .
- 2) Calculate the RMS of these vibration samples, then the eigenvalue vector $\mathbf{X} = [x_1, x_2, \dots, x_n]^T$ with the size of $n \times 1$ can be obtained. Select the eigenvalue vector $\mathbf{S} = [x_1, x_2, \dots, x_m]^T$ with m samples as the training vector.
- 3) Normalize the vector $\mathbf{S}$ by linear normalization and get the normalized vector $\mathbf{Y} = [y_1, y_2, \dots, y_m]^T$.

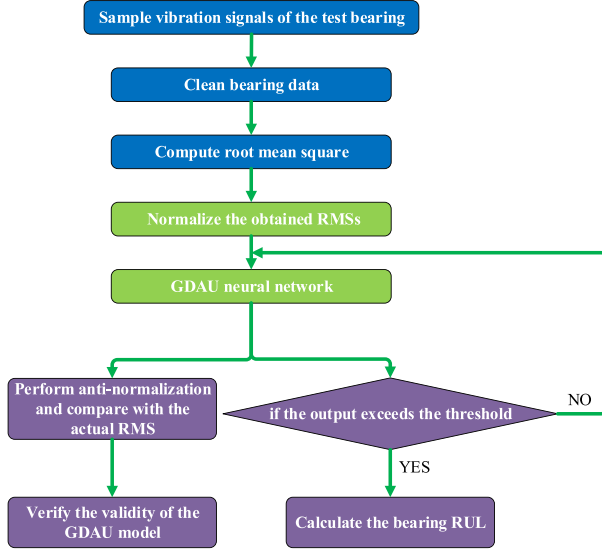


Fig. 6. Flowchart of bearing RUL prediction approach.

4) Reconstruct matrix $\mathbf{W}$

$$\mathbf{W} = \begin{bmatrix} y_1 & y_2 & \cdots & y_{m-k} \\ y_2 & y_3 & \cdots & y_{m-k+1} \\ \vdots & \vdots & \ddots & \vdots \\ y_{k+1} & y_{k+2} & \cdots & y_m \end{bmatrix} = \begin{bmatrix} \mathbf{w}_1 \\ \mathbf{w}_2 \\ \vdots \\ \mathbf{w}_{k+1} \end{bmatrix} \quad (16)$$

where k is the cell number of input layer and

$$\mathbf{w}_i = [y_i \ y_{i+1} \ \cdots \ y_{m-k+i-1}] \quad (17)$$

- 5) The first k vectors of the matrix $\mathbf{W}$ are set as the input of the GDAU neural network and the last vector $\mathbf{w}_{k+1}$ is regarded as the output, and then the GDAU neural network is trained.
- 6) The trained GDAU neural network is defined as the mapping function f . By inputting the last k vectors of the matrix $\mathbf{W}$ into the trained GDAU neural network, the output $\bar{\mathbf{w}}_{k+2}$ at the first prediction time (FPT) is calculated as

$$\bar{\mathbf{w}}_{k+2} = f(\mathbf{w}_2, \mathbf{w}_3, \dots, \mathbf{w}_{k+1}) \quad (18)$$

Then, the matrix $\mathbf{W}$ is updated by

$$\mathbf{W} = [\mathbf{w}_1 \ \mathbf{w}_2 \ \cdots \ \mathbf{w}_{k+1} \ \bar{\mathbf{w}}_{k+2}]^T \quad (19)$$

- 7) Repeat step 6 several times, the successive predicted vectors $\bar{\mathbf{w}}_i$ are obtained as

$$\begin{aligned} \bar{\mathbf{w}}_{k+3} &= f(\mathbf{w}_3, \mathbf{w}_4, \dots, \mathbf{w}_{k+1}, \bar{\mathbf{w}}_{k+2}) \\ \bar{\mathbf{w}}_{k+4} &= f(\mathbf{w}_4, \mathbf{w}_5, \dots, \mathbf{w}_{k+1}, \bar{\mathbf{w}}_{k+2}, \bar{\mathbf{w}}_{k+3}) \\ \bar{\mathbf{w}}_{k+5} &= f(\mathbf{w}_5, \mathbf{w}_6, \dots, \mathbf{w}_{k+1}, \bar{\mathbf{w}}_{k+2}, \bar{\mathbf{w}}_{k+3}, \bar{\mathbf{w}}_{k+4}) \\ &\vdots \\ \bar{\mathbf{w}}_i &= f(\mathbf{w}_{i-k}, \mathbf{w}_{i-k+1}, \dots, \mathbf{w}_{k+1}, \bar{\mathbf{w}}_{k+2}, \dots, \\ &\quad \bar{\mathbf{w}}_{k+(i-k-2)}, \bar{\mathbf{w}}_{k+(i-k-1)}) \end{aligned} \quad (20)$$

When the antinormalized value of the predicted data $\bar{y}_{m-k+i-1} (\bar{y}_{m-k+i-1} \in \bar{\mathbf{w}}_i)$ exceeds the preset threshold, the

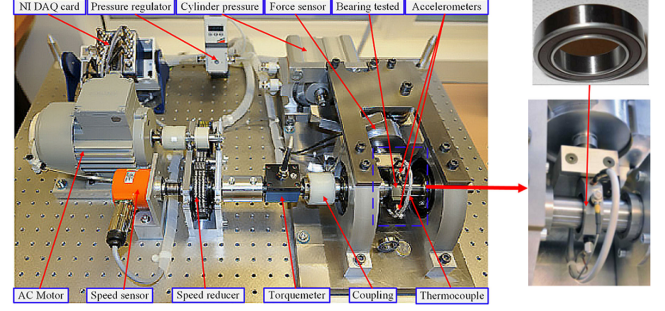


Fig. 7. PRONOSTIA test rig.

bearing remaining life is calculated as $(i - m) \times T_s$. Then the vector $\bar{\mathbf{x}} = (\bar{x}_{m+1}, \bar{x}_{m+2}, \dots, \bar{x}_n)^T$ is obtained by antinormalizing the predicted data series, which will be compared with the actual data $\mathbf{x}' = (x_{m+1}, x_{m+2}, \dots, x_n)^T$ to verify the validity of the proposed approach.

IV. EXPERIMENTAL RESULTS

A. Data Set Description

The experimental data sets come from the PHM 2012 challenge data sets, which are obtained by PRONOSTIA test rig. This test rig mainly contains driven motor, various sensors, tested rolling ball bearing, and loading device, which is shown in Fig. 7. The dimensions of tested bearings are given below: the diameter of rolling elements is 3.5 mm; the number of rolling elements is 13; the diameter of the outer race is 29.1 mm; the diameter of inner race is 22.1 mm, and the mean diameter of bearing is 25.6 mm. Two synthetic rubber seals are applied to prevent leakage of lubricant as well as entry of dust, water, and other harmful material. Two accelerometers were used to sample horizontal and vertical vibration signals, respectively. The sampling frequency was 25.6 kHz and the sampling time length was 0.1 s, which was recorded per 10 s. More details about the test rig and the challenge data sets can be found in [37].

In the experiment, the run-to-failure bearing1_1 and the bearing1_3 are used to validate the proposed RUL prediction approach, and the working condition is rotating speed is 1800 r/min and loading force is 4000 N. The bearing1_1 and bearing1_3, respectively, contain 2803 and 2375 samples. Fig. 8(a) and (b) shows the life-cycle waveforms of the bearing1_1 and bearing1_3 data sets, respectively.

B. Evaluation Index

In order to compare the performance of the GDAU neural network with GRU and its variants, some appropriate evaluation indexes are required. A score function is used in the IEEE PHM 2012 challenge [37] that can assess the predictive performance, and it is computed by

$$Er_i = \frac{Rul_i - \overline{Rul}_i}{Rul_i} \times 100 \quad (21)$$

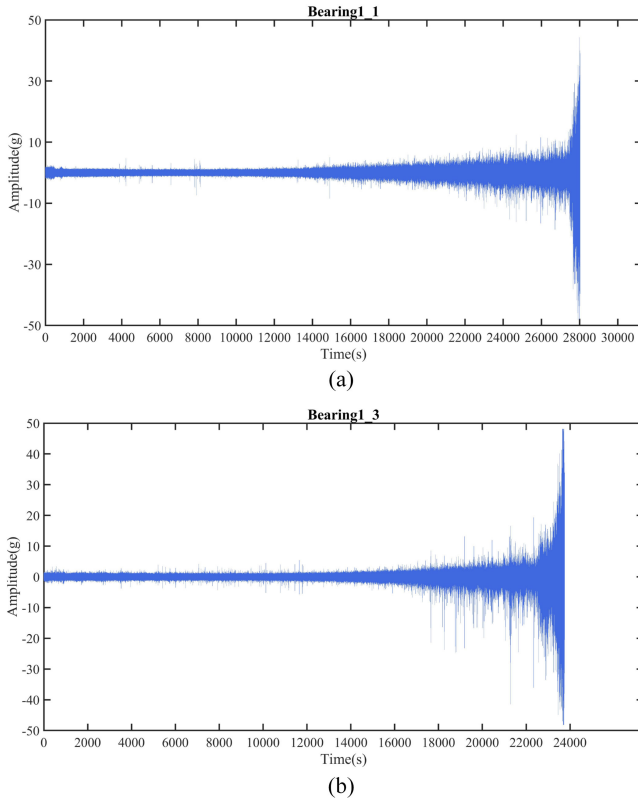


Fig. 8. Life-cycle waveforms of two bearing data sets. (a) bearing1_1 data set. (b) bearing1_3 data set.

$$A_i = \begin{cases} \exp(-\ln(0.5) \cdot (Er_i/5)), & Er_i \leq 0 \\ \exp(\ln(0.5) \cdot (Er_i/20)), & Er_i > 0 \end{cases} \quad (22)$$

$$\text{Score} = \frac{1}{20} \sum_{i=1}^{20} A_i \quad (23)$$

where Rul and $\overline{Rul}$ represent the actual RUL and estimated RUL, respectively; Er_i denotes the percent error for the i th experiment; A_i is the score of accuracy in the i th experiment; and the final score is the mean of all experiment's scores.

Except for the score function, mean absolute error (MAE), normalized root mean square error (NRMSE), mean absolute percentage error (MAPE) and root mean square error (RMSE) are used to evaluate the accuracy of predictions. They are, respectively, computed as follows:

$$\text{MAE} = \frac{1}{20} \sum_{i=1}^{20} |Rul_i - \overline{Rul}_i| \quad (24)$$

$$\text{NRMSE} = \frac{\sqrt{\frac{1}{20} \sum_{i=1}^{20} (Rul_i - \overline{Rul}_i)^2}}{\frac{1}{20} \sum_{i=1}^{20} \overline{Rul}_i} \quad (25)$$

$$\text{MAPE} = \frac{1}{20} \sum_{i=1}^{20} \left| \frac{Rul_i - \overline{Rul}_i}{Rul_i} \right| \times 100\% \quad (26)$$

$$\text{RMSE} = \sqrt{\frac{1}{20} \sum_{i=1}^{20} (Rul_i - \overline{Rul}_i)^2} \quad (27)$$

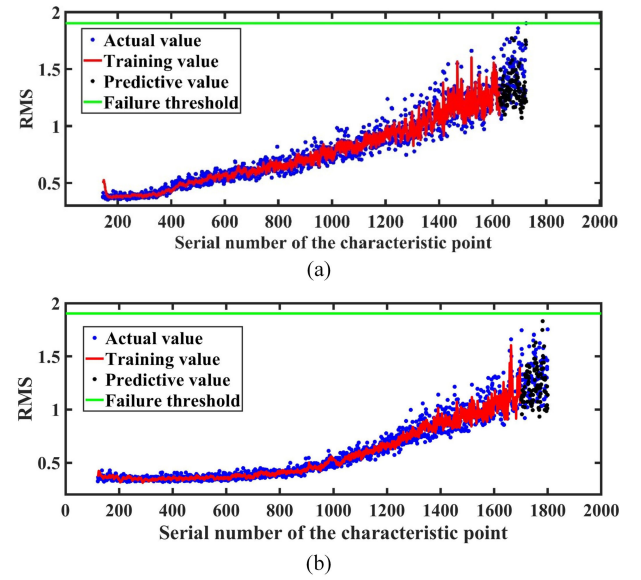


Fig. 9. Training curves, predicted curves, and actual curve for 100 predictive points. (a) bearing1_1 data set. (b) bearing1_3 data set.

Via the above evaluation indexes, the performance of GDAU neural network and other current neural networks can be comprehensively evaluated and compared.

C. RUL Prediction and Comparison

The RMS vectors of bearing1_1 and bearing1_3 data sets are first calculated, and then they are applied to RUL prediction. For the bearing1_1 dataset, the first 1000 health characteristic points acquired under the steady stage are little helpful to predict RUL, thus a total of 1725 health characteristic points from 1001 to 2725 are selected for RUL prediction. The first 1625 points are set as the training set to predict the next 100 points. Similarly, for the bearing1_3 data set, 1800 characteristic points from 439 to 2238 are chosen for RUL prediction, and the first 1700 points are trained to predict the last 100 points.

The values of RMS are able to describe the tendency of bearing degradation reasonably. In early stage, the value of RMS is small due to the fact that there is no fault occurred or the fault is slight. As time goes on, the fault becomes more and more severe, which results in that the value of RMS become larger. Finally, we find from the experiment that the value of RMS reached the maximum value 1.903 when the tested bearing loses efficacy, so it is selected as the failure threshold in this work. With the GDAU neural network, the number of input layer, hidden layer and output layer of bearing1_1 are, respectively, set as 144, 22, and 1. For bearing1_3, they are set as 120, 21, and 1. The learning rate of bearing1_1 and bearing1_3 are 0.08 and 0.06. The training and predictive curves of the two bearing data sets are, respectively, shown in Fig. 9(a) and (b). From this figure, it is easy to note that both training and predicted values of RMS are close to the actual values, and particularly the training and predictive curves have similar trend with the real degenerated curve.

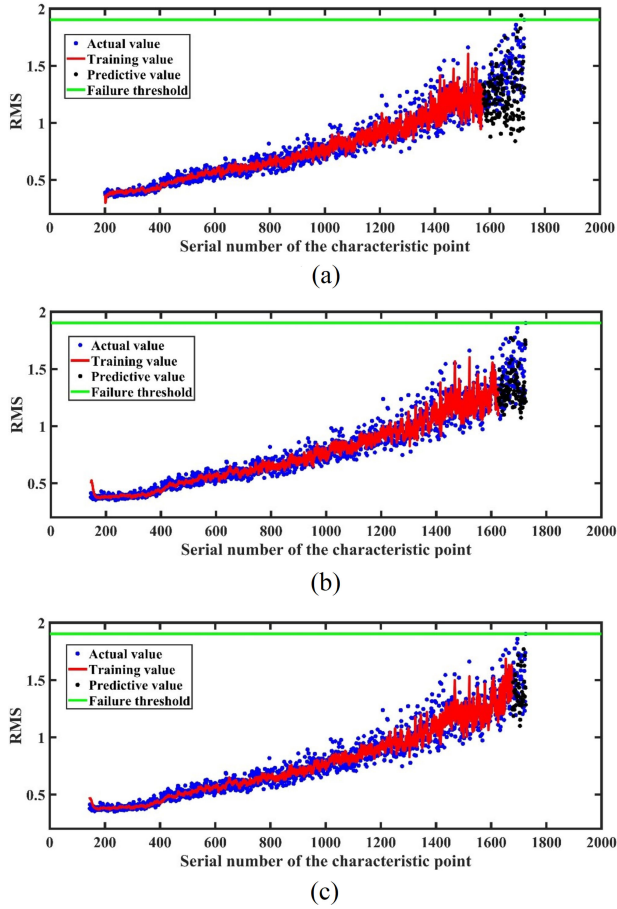


Fig. 10. Predictive results of bearing1_1 data set for different amounts of predicted points. (a) 150. (b) 100. (c) 50.

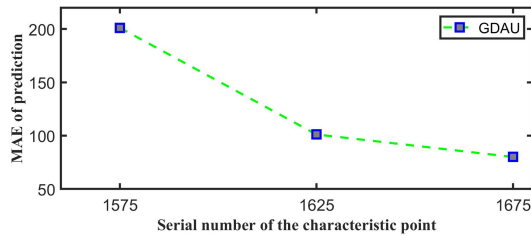


Fig. 11. MAEs for different predictive points.

To demonstrate the long-term and short-term prediction capabilities of GDAU, we use the bearing1_1 data set, and separately set the number of predicted points as 150, 100, and 50. Fine-tuning the number of input layer and learning rate and the predictive results are shown in Fig. 10. Evidently, as the number of predictive points decreases, the predicted value is getting closer to the true value of RMS. In three examples, the predicted curve and the true curve of RMS have the similar trend, so the RUL can be well estimated.

Next, we calculate the MAEs for different predictive points to quantitatively assess the predictive ability of GDAU, and the computed result is shown in Fig. 11. It can be seen from this figure that MAE decreases with the increase of the number of the known health characteristic points. Moreover, the percent

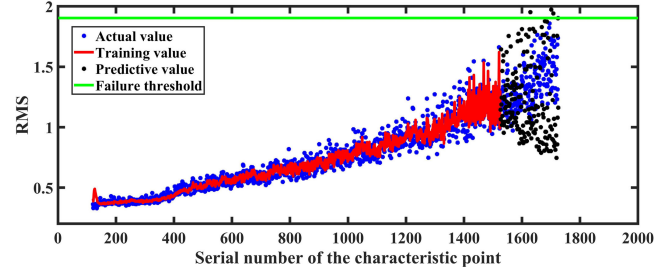


Fig. 12. Training curve, predicted curve, and actual curve for the bearing1_1 data set with 1525 training samples.

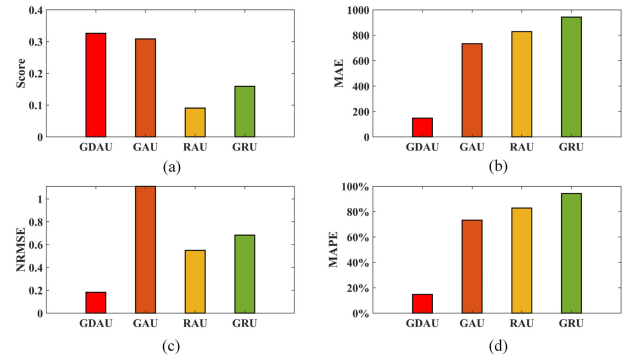


Fig. 13. Evaluation indexes for different neural networks. (a) score. (b) MAE. (c) NRMSE. (d) MAPE.

error of RUL prediction is calculated as 13.4% when the number of predictive points is 150. It indicates that the GDAU has the long-term predictive capability and the prediction accuracy is basically satisfying. In order to further demonstrate the long-term predictive capability of GDAU, we also use the bearing1_1 data set to predict 200 health characteristic points, i.e., 1525 bearing samples are known. The obtained result is illustrated in Fig. 12. By calculation, MAE is 667 and the prediction accuracy is 66.7% and the percent error of RUL prediction is 33.3%. It then follows that GDAU still has certain prediction capability even though there is a smaller number of known samples.

For comparison, GRU, RAU, GAU, and GDAU are separately employed to predict the bearing RULs with two bearing data sets. To assure the fairness and robustness of the comparative results, each type of neural network uses optimized hyper parameters and implements prediction on each bearing data set for 10 times. In the comparative experiment, 100 health characteristic points are predicted, and then the score function, MAE, NRMSE, and MAPE obtained by each method are, respectively, computed. The average results of these indexes are illustrated in Fig. 13. From this figure, it can be obviously noted that GDAU has the highest score and the lowest error among these neural networks. Especially for MAE, NRMSE, and MAPE, GDAU is greatly superior to GRU and its variants. This is mainly because that RAU and GDAU neural networks only apply attention mechanism from a single aspect, whereas the proposed GDAU neural network makes full use of the contribution of input data and intermediate results on the predicted data. We can also know from Fig. 13 that the score of RAU is much lower than that

TABLE I
RUL PREDICTION COMPARISON

Indexes	GDAU	MMALSTM	DGAN	DNN
MAE	148	196	209	234
RMSE	0.187	0.237	0.248	0.330
MAPE	14.8	28.6	43.9	116.7

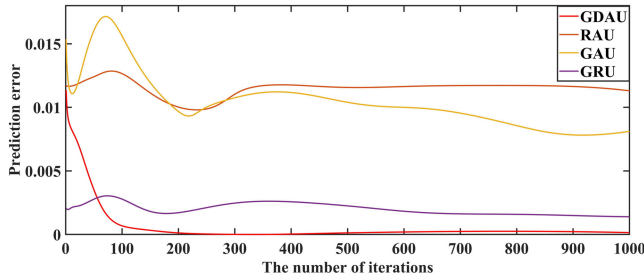


Fig. 14. Prediction error curves of different neural networks.

of GRU even though MAE, NRMSE and MAPE of RAU are smaller than those of GRU. This indicates that just applying attention mechanism on the information of the input data and hidden state is less sufficient to improve the prediction ability of GRU. Moreover, GAU has higher comprehensive predictive performance than RAU and GRU, and it also proves that the newly designed attention gate in GAU is more helpful for capturing dependencies than that in RAU. By integrating two attention gates, GDAU achieves the best predictive performance. From the above comparative results, it can be concluded that the proposed GDAU-based RUL prediction method has higher prediction accuracy than the methods based on other types of GRU.

To further illustrate the advantage of the GDAU neural network, MMALSTM [33], DGAN [15], and DNN [17] are used for comparison. With the same data sets, the indexes for evaluating these methods are calculated and listed in Table I. It can be easily noted from Table I that the GDAU neural network achieves the best performance among these methods. In particular, MAPE is greatly superior to other neural networks. The comparative results further indicate that the GDAU neural network has great potential in RUL prediction owing to its ability to make full use of input data and intermediate information by two attention gates.

Except for the prediction accuracy, the convergence speeds of four approaches are compared. Similarly, the bearing 1_1 data set is used for comparison. The curves of prediction error obtained by GRU, RAU, GAU, and GDAU are shown in Fig. 14. It can be seen from this figure that the prediction error of GDAU becomes steady after 200 iterations, while the prediction error curves of GRU, RAU, and GAU do not converge even after 1000 iterations. Since two attention gates can simultaneously pay attention to the crucial information of input data, the hidden state at the previous time instant, and the outputs of the reset and update gates, the GDAU neural network has faster convergence speed than other neural networks. Moreover, we can note from Fig. 14 that the prediction error of GDAU neural network is lower than other neural networks. It again verifies that GDAU has higher accuracy in the prediction of bearing RUL.

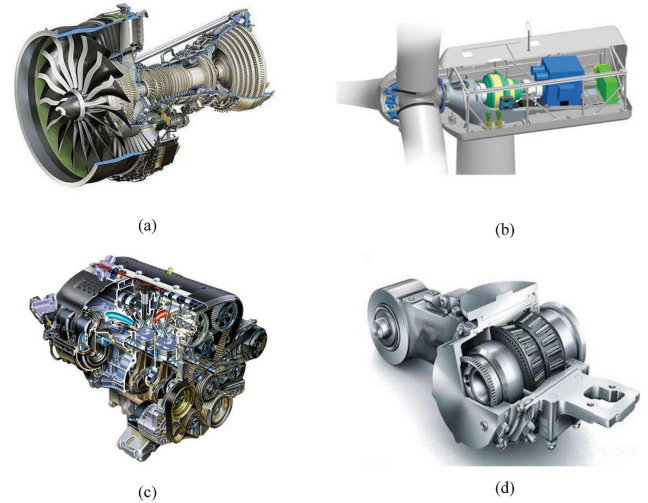


Fig. 15. Industrial applications of the GDAU neural network. (a) Aero engine. (b) Wind turbine. (c) Automobile engine. (d) High-speed rail.

D. Discussion

The proposed GDAU neural network can be used to predict the bearing RUL for aero engine, wind turbine, automobile engine, and high-speed rail. The industrial applications of GDAU neural network are shown in Fig. 15.

V. CONCLUSION

This article explored a novel RUL prediction approach for rolling bearings. A new attention gate was designed to extract the important information from the outputs of the reset and update gates, and then GAU was constructed based on this attention gate. By integrating the new attention gate with the attention gate used in RAU, GDAU was proposed, which could improve the prediction accuracy and convergence speed. Then, using RMS as the HI, a GDAU-based RUL prediction method for rolling bearings was proposed. Via the PHM 2012 challenge data sets, the performance of the proposed method was verified and compared to the methods based on GRU, RAU, GAU, MMALSTM, DGAN, and DNN. The comparative results demonstrate that the GDAU-based method possesses the stronger long-term and short-term prediction ability, so it is more suitable for bearing RUL prediction. The code is available for download online.¹

The main goal of this article was to design an advanced neural network for predicting bearing RUL, whereas the network structure can be further optimized in the future study. Moreover, it is worth exploring the bearing RUL prediction under transient (nonstationary) operating conditions, and the HI to better describe the tendency of bearing degradation need be researched.

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¹[Online]. Available: [https://github.com/QinYi-team/Code/tree/master/Gated%20dual%20attention%20unit%20\(GDAU\)](https://github.com/QinYi-team/Code/tree/master/Gated%20dual%20attention%20unit%20(GDAU)).

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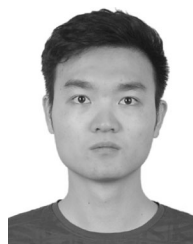
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